CLASS 10 · CBSE · NCERT MATHEMATICS · NCERT SOLUTIONS

Chapter 5: Arithmetic Progressions

Step-by-step solutions to the textbook exercises: 49 questions

Questions are numbered as in the textbook. Try each one yourself before reading the solution.

1 Exercise 5.1

Q1 In which of these situations does the list of numbers form an AP, and why? (i) The taxi fare after each km, when the fare is ₹15 for the first km and ₹8 for each additional km. (ii) The amount of air left in a cylinder when a pump removes $\frac{1}{4}$ of the remaining air each time. (iii) The cost of digging a well after each metre, when it costs ₹150 for the first metre and ₹50 more for each further metre. (iv) The amount in an account each year, when ₹10,000 is deposited at 8% compound interest per annum.

SOLUTION

A list is an AP when each term is obtained by adding the same fixed number to the term before it.

(i) The fares are 15, 23, 31, 39, ... Each is 8 more than the one before, so it is an AP.

(ii) If the cylinder holds V units of air, the amounts left are V, $\frac{3V}{4}$, $\frac{9V}{16}$, ... The differences $-\frac{V}{4}$ and $-\frac{3V}{16}$ are not equal, so it is not an AP.

(iii) The costs are 150, 200, 250, 300, ... Each is 50 more than the one before, so it is an AP.

(iv) The amounts are $10000, 10000(1.08), 10000(1.08)^2$, ... that is 10000, 10800, 11664, ... The differences 800 and 864 are not equal, so it is not an AP.

Answer: (i) and (iii) form APs. (ii) and (iv) do not.

Q2 Write the first four terms of the AP when (i) $a = 10, d = 10$ (ii) $a = -2, d = 0$ (iii) $a = 4, d = -3$ (iv) $a = -1, d = \frac{1}{2}$ (v) $a = -1.25, d = -0.25$.

SOLUTION

Start with a and keep adding d.

Answer: (i) $10, 20, 30, 40$ **(ii)** $-2, -2, -2, -2$ **(iii)** $4, 1, -2, -5$ **(iv)** $-1, -\frac{1}{2}, 0, \frac{1}{2}$ **(v)** $-1.25, -1.50, -1.75, -2.00$

Q3 Identify the first term a and the common difference d of each progression: (i) $3, 1, -1, -3, \ldots$ (ii) $-5, -1, 3, 7, \ldots$ (iii) $\frac{1}{3}, \frac{5}{3}, \frac{9}{3}, \frac{13}{3}, \ldots$ (iv) $0.6, 1.7, 2.8, 3.9, \ldots$

SOLUTION

The common difference is the second term minus the first.

Answer: (i) $a = 3, d = -2$ **(ii)** $a = -5, d = 4$ **(iii)** $a = \frac{1}{3}, d = \frac{4}{3}$ **(iv)** $a = 0.6, d = 1.1$

Q4 Which of the following are APs? For each AP, find d and write three more terms. (i) $2, 4, 8, 16, \ldots$ (ii) $2, \frac{5}{2}, 3, \frac{7}{2}, \ldots$ (iii) $-1.2, -3.2, -5.2, -7.2, \ldots$ (iv) $-10, -6, -2, 2, \ldots$ (v) $3, 3+\sqrt{2}, 3+2\sqrt{2}, 3+3\sqrt{2}, \ldots$ (vi) $0.2, 0.22, 0.222, 0.2222, \ldots$ (vii) $0, -4, -8, -12, \ldots$ (viii) $-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}, \ldots$ (ix) $1, 3, 9, 27, \ldots$ (x) $a, 2a, 3a, 4a, \ldots$ (xi) $a, a^2, a^3, a^4, \ldots$ (xii) $\sqrt{2}, \sqrt{8}, \sqrt{18}, \sqrt{32}, \ldots$ (xiii) $\sqrt{3}, \sqrt{6}, \sqrt{9}, \sqrt{12}, \ldots$ (xiv) $1^2, 3^2, 5^2, 7^2, \ldots$ (xv) $1^2, 5^2, 7^2, 73, \ldots$

SOLUTION

Check whether the differences between consecutive terms are all equal.

Part	Differences	AP?	d and the next three terms
(i)	2, 4, 8	No	
(ii)	$\frac{1}{2}, \frac{1}{2}, \frac{1}{2}$	Yes	$d = \frac{1}{2}$: $4, \frac{9}{2}, 5$
(iii)	$-2, -2, -2$	Yes	$d = -2$: $-9.2, -11.2, -13.2$
(iv)	4, 4, 4	Yes	$d = 4$: $6, 10, 14$
(v)	$\sqrt{2}, \sqrt{2}, \sqrt{2}$	Yes	$d = \sqrt{2}$: $3+4\sqrt{2}, 3+5\sqrt{2}, 3+6\sqrt{2}$
(vi)	0.02, 0.002	No	
(vii)	$-4, -4, -4$	Yes	$d = -4$: $-16, -20, -24$
(viii)	0, 0, 0	Yes	$d = 0$: $-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}$
(ix)	2, 6, 18	No	
(x)	a, a, a	Yes	$d = a$: $5a, 6a, 7a$
(xi)	$a^2 - a, a^3 - a^2$	No (in general)	
(xii)	The terms are $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, 4\sqrt{2}$	Yes	$d = \sqrt{2}$: $\sqrt{50}, \sqrt{72}, \sqrt{98}$
(xiii)	$\sqrt{6} - \sqrt{3}, 3 - \sqrt{6}$	No	
(xiv)	The terms are 1, 9, 25, 49: differences 8, 16, 24	No	
(xv)	The terms are 1, 25, 49, 73: differences 24, 24, 24	Yes	$d = 24$: $97, 121, 145$

Answer: (ii), (iii), (iv), (v), (vii), (viii), (x), (xii) and (xv) are APs. The others are not.

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2 Exercise 5.2

Q1 Fill in the blanks, where a is the first term, d the common difference and a_n the nth term of an AP.

	a	d	n	a_n
(i)	7	3	8	?
(ii)	-18	?	10	0
(iii)	?	-3	18	-5
(iv)	-18.9	2.5	?	3.6
(v)	3.5	0	105	?

SOLUTION

Use $a_n = a + (n-1)d$ in each row.

(i) $a_8 = 7 + 7 \times 3 = 28$.

(ii) $0 = -18 + 9d$, so $d = 2$.

(iii) $-5 = a + 17 \times (-3)$, so $a = 46$.

(iv) $3.6 = -18.9 + (n-1) \times 2.5$, so $n - 1 = \dfrac{22.5}{2.5} = 9$ and $n = 10$.

(v) $a_{105} = 3.5 + 104 \times 0 = 3.5$.

Answer: (i) $a_n = 28$ (ii) $d = 2$ (iii) $a = 46$ (iv) $n = 10$ (v) $a_n = 3.5$

Q2 Choose the correct option and justify it. (i) The 30th term of the AP $10, 7, 4, \ldots$ is (A) 97 (B) 77 (C) -77 (D) -87. (ii) The 11th term of the AP $-3, -\frac{1}{2}, 2, \ldots$ is (A) 28 (B) 22 (C) -38 (D) $-48\frac{1}{2}$.

SOLUTION

(i) $a = 10$, $d = -3$: $a_{30} = 10 + 29 \times (-3) = -77$.

(ii) $a = -3$, $d = -\dfrac{1}{2} + 3 = \dfrac{5}{2}$: $a_{11} = -3 + 10 \times \dfrac{5}{2} = 22$.

Answer: (i) (C) -77 (ii) (B) 22

Q3 Find the missing terms (shown as $\square$) in these APs: (i) $2, \square, 26$ (ii) $\square, 13, \square, 3$ (iii) $5, \square, \square, 9\frac{1}{2}$ (iv) $-4, \square, \square, \square, \square, 6$ (v) $\square, 38, \square, \square, \square, -22$.

SOLUTION

(i) The middle term is the average of its neighbours: $\dfrac{2 + 26}{2} = 14$.

(ii) $a + d = 13$ and $a + 3d = 3$, so $2d = -10$, $d = -5$ and $a = 18$. The third term is $13 - 5 = 8$.

(iii) $a = 5$ and $a + 3d = \dfrac{19}{2}$, so $d = \dfrac{3}{2}$. The terms are $\dfrac{13}{2}$ and 8.

(iv) $a = -4$ and $a + 5d = 6$, so $d = 2$. The terms are $-2, 0, 2, 4$.

(v) $a + d = 38$ and $a + 5d = -22$, so $4d = -60$, $d = -15$ and $a = 53$. The AP is $53, 38, 23, 8, -7, -22$.

Answer: (i) 14 (ii) 18 and 8 (iii) $6\frac{1}{2}$ and 8 (iv) $-2, 0, 2, 4$ (v) 53, then 23, 8, -7

Q4 Which term of the AP $3, 8, 13, 18, \ldots$ is 78?

SOLUTION

$a = 3, d = 5$: $3 + (n-1) \times 5 = 78$ gives $n - 1 = 15$.

Answer: the 16th term

Q5 Find the number of terms in each AP: (i) $7, 13, 19, \ldots, 205$ (ii) $18, 15\frac{1}{2}, 13, \ldots, -47$.

SOLUTION

(i) $a = 7, d = 6$: $7 + (n-1) \times 6 = 205$ gives $n - 1 = 33$, so $n = 34$.

(ii) $a = 18, d = -\frac{5}{2}$: $18 - \frac{5}{2}(n-1) = -47$ gives $n - 1 = 26$, so $n = 27$.

Answer: (i) 34 (ii) 27

Q6 Is -150 a term of the AP $11, 8, 5, 2, \ldots$?

SOLUTION

$a = 11, d = -3$: $11 - 3(n-1) = -150$ gives $n - 1 = \frac{161}{3}$, which is not a whole number.

Answer: no, -150 is not a term of this AP

Q7 An AP has 38 as its 11th term and 73 as its 16th term. What is its 31st term?

SOLUTION

$a + 10d = 38$ and $a + 15d = 73$. Subtracting, $5d = 35$, so $d = 7$ and $a = -32$.

$a_{31} = -32 + 30 \times 7 = 178$.

Answer: 178

Q8 An AP has 50 terms. Its 3rd term is 12 and its last term is 106. Find the 29th term.

SOLUTION

$a + 2d = 12$ and $a + 49d = 106$. Subtracting, $47d = 94$, so $d = 2$ and $a = 8$.

$a_{29} = 8 + 28 \times 2 = 64$.

Answer: 64

Q9 The 3rd and 9th terms of an AP are 4 and -8. Which term is zero?

SOLUTION

$a + 2d = 4$ and $a + 8d = -8$. Subtracting, $6d = -12$, so $d = -2$ and $a = 8$.

$8 - 2(n-1) = 0$ gives $n = 5$.

Answer: the 5th term

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Q10 In an AP, the 17th term is 7 more than the 10th term. What is the common difference?

SOLUTION

$(a + 16d) - (a + 9d) = 7$ gives $7d = 7$.

Answer: $d = 1$

Q11 Which term of the AP $3, 15, 27, 39, \ldots$ is 132 more than its 54th term?

SOLUTION

$d = 12$. Each step adds 12, and $132 \div 12 = 11$, so the required term is 11 places after the 54th.

Answer: the 65th term

Q12 Two APs have the same common difference. Their 100th terms differ by 100. By how much do their 1000th terms differ?

SOLUTION

Let the first terms be a and b and the common difference be d. The difference of the nth terms is $[a + (n-1)d] - [b + (n-1)d] = a - b$, which does not depend on n.

So $a - b = 100$ for every pair of matching terms.

Answer: 100

Q13 Count the three-digit numbers that are multiples of 7.

SOLUTION

They are $105, 112, \ldots, 994$, an AP with $d = 7$.

$105 + (n-1) \times 7 = 994$ gives $n - 1 = 127$.

Answer: 128

Q14 Count the multiples of 4 that lie strictly between 10 and 250.

SOLUTION

They are $12, 16, \ldots, 248$: $12 + (n-1) \times 4 = 248$ gives $n - 1 = 59$.

Answer: 60

Q15 For what value of n are the nth terms of the APs $63, 65, 67, \ldots$ and $3, 10, 17, \ldots$ equal?

SOLUTION

$63 + 2(n-1) = 3 + 7(n-1)$ gives $60 = 5(n-1)$, so $n - 1 = 12$.

Answer: $n = 13$

MathSetu

Q16 Find the AP whose third term is 16 and whose 7th term exceeds the 5th term by 12.

SOLUTION

$a_7 - a_5 = 2d = 12$, so $d = 6$. Then $a + 2d = 16$ gives $a = 4$.

Answer: $4, 10, 16, 22, \ldots$

Q17 Counting backwards from the last term, find the 20th term of the AP $3, 8, 13, \ldots, 253$.

SOLUTION

Read the AP backwards: $253, 248, 243, \ldots$ with first term 253 and common difference -5.

Its 20th term is $253 + 19 \times (-5) = 158$.

Answer: 158

Q18 In an AP, the 4th and 8th terms add up to 24, and the 6th and 10th terms add up to 44. Write down the first three terms.

SOLUTION

$(a + 3d) + (a + 7d) = 24$ gives $a + 5d = 12$. $(a + 5d) + (a + 9d) = 44$ gives $a + 7d = 22$.

Subtracting, $2d = 10$, so $d = 5$ and $a = -13$.

Answer: $-13, -8, -3$

Q19 A worker joined in 1995 on a yearly pay of ₹5000, with a rise of ₹200 every year. In which year did the pay reach ₹7000?

SOLUTION

The salaries form an AP with $a = 5000$ and $d = 200$: $5000 + (n - 1) \times 200 = 7000$ gives $n = 11$.

The 11th year of work is $1995 + 10 = 2005$.

Answer: in 2005 (the 11th year)

Q20 A person saved ₹5 in the first week of a year and increased the weekly saving by ₹1.75 each week. In the nth week the saving was ₹20.75. Find n.

SOLUTION

$5 + (n - 1) \times 1.75 = 20.75$ gives $n - 1 = \dfrac{15.75}{1.75} = 9$.

Answer: $n = 10$

MathSetu

3 Exercise 5.3

Q1 Find the sum of each AP: (i) $2, 7, 12, \ldots$ to 10 terms (ii) $-37, -33, -29, \ldots$ to 12 terms (iii) $0.6, 1.7, 2.8, \ldots$ to 100 terms (iv) $\frac{1}{15}, \frac{1}{12}, \frac{1}{10}, \ldots$ to 11 terms.

SOLUTION

Use $S_n = \frac{n}{2}[2a + (n-1)d]$.

(i) $S_{10} = 5[4 + 9 \times 5] = 245.$

(ii) $S_{12} = 6[-74 + 11 \times 4] = 6 \times (-30) = -180.$

(iii) $S_{100} = 50[1.2 + 99 \times 1.1] = 50 \times 110.1 = 5505.$

(iv) $d = \frac{1}{12} - \frac{1}{15} = \frac{1}{60}.\ S_{11} = \frac{11}{2}\left[\frac{2}{15} + \frac{10}{60}\right] = \frac{11}{2} \times \frac{18}{60} = \frac{33}{20}.$

Answer: (i) 245 (ii) -180 (iii) 5505 (iv) $\frac{33}{20}$

Q2 Find the sums: (i) $7 + 10\frac{1}{2} + 14 + \cdots + 84$ (ii) $34 + 32 + 30 + \cdots + 10$ (iii) $-5 + (-8) + (-11) + \cdots + (-230).$

SOLUTION

First find the number of terms from $l = a + (n-1)d$, then use $S_n = \frac{n}{2}(a + l)$.

(i) $d = 3.5$: $n = \frac{84 - 7}{3.5} + 1 = 23.\ S = \frac{23}{2}(7 + 84) = 1046.5.$

(ii) $d = -2$: $n = \frac{10 - 34}{-2} + 1 = 13.\ S = \frac{13}{2}(34 + 10) = 286.$

(iii) $d = -3$: $n = \frac{-230 + 5}{-3} + 1 = 76.\ S = \frac{76}{2}(-5 - 230) = -8930.$

Answer: (i) $1046\frac{1}{2}$ (ii) 286 (iii) -8930

Q3 In an AP: (i) $a = 5$, $d = 3$, $a_n = 50$: find n and S_n. (ii) $a = 7$, $a_{13} = 35$: find d and S_{13}. (iii) $a_{12} = 37$, $d = 3$: find a and S_{12}. (iv) $a_3 = 15$, $S_{10} = 125$: find d and a_{10}. (v) $d = 5$, $S_9 = 75$: find a and a_9. (vi) $a = 2$, $d = 8$, $S_n = 90$: find n and a_n. (vii) $a = 8$, $a_n = 62$, $S_n = 210$: find n and d. (viii) $a_n = 4$, $d = 2$, $S_n = -14$: find n and a. (ix) $a = 3$, $n = 8$, $S = 192$: find d. (x) $l = 28$, $S = 144$ and there are 9 terms: find a.

SOLUTION

(i) $5 + 3(n-1) = 50$ gives $n = 16$. $S_{16} = \frac{16}{2}(5 + 50) = 440$.

(ii) $7 + 12d = 35$ gives $d = \frac{7}{3}$. $S_{13} = \frac{13}{2}(7 + 35) = 273$.

(iii) $a + 11 \times 3 = 37$ gives $a = 4$. $S_{12} = \frac{12}{2}(4 + 37) = 246$.

(iv) $a + 2d = 15$ and $5(2a + 9d) = 125$, that is $2a + 9d = 25$. Putting $a = 15 - 2d$: $30 + 5d = 25$, so $d = -1$ and $a = 17$. $a_{10} = 17 - 9 = 8$.

(v) $\frac{9}{2}(2a + 40) = 75$ gives $2a + 40 = \frac{50}{3}$, so $a = -\frac{35}{3}$. $a_9 = -\frac{35}{3} + 40 = \frac{85}{3}$.

(vi) $\frac{n}{2}[4 + 8(n-1)] = 90$ gives $2n^2 - n - 45 = 0$, so $(2n + 9)(n - 5) = 0$ and $n = 5$. $a_5 = 2 + 32 = 34$.

(vii) $\frac{n}{2}(8 + 62) = 210$ gives $n = 6$. $8 + 5d = 62$ gives $d = \frac{54}{5}$.

(viii) $a = 4 - 2(n-1) = 6 - 2n$. $\frac{n}{2}(6 - 2n + 4) = -14$ gives $n^2 - 5n - 14 = 0$, so $(n - 7)(n + 2) = 0$ and $n = 7$. Then $a = -8$.

(ix) $\frac{8}{2}(6 + 7d) = 192$ gives $6 + 7d = 48$, so $d = 6$.

(x) $\frac{9}{2}(a + 28) = 144$ gives $a + 28 = 32$, so $a = 4$.

Answer: (i) $n = 16$, $S_n = 440$ **(ii)** $d = \frac{7}{3}$, $S_{13} = 273$ **(iii)** $a = 4$, $S_{12} = 246$ **(iv)** $d = -1$, $a_{10} = 8$ **(v)** $a = -\frac{35}{3}$, $a_9 = \frac{85}{3}$ **(vi)** $n = 5$, $a_n = 34$ **(vii)** $n = 6$, $d = \frac{54}{5}$ **(viii)** $n = 7$, $a = -8$ **(ix)** $d = 6$ **(x)** $a = 4$

Q4 How many terms of the AP $9, 17, 25, \ldots$ have to be added to get 636?

SOLUTION

$\frac{n}{2}[18 + 8(n-1)] = 636$ gives $4n^2 + 5n - 636 = 0$.

$D = 25 + 10176 = 10201 = 101^2$, so $n = \frac{-5 + 101}{8} = 12$ (the other root is negative).

Answer: 12 terms

Q5 An AP begins with 5 and ends with 45, and its terms add up to 400. Find how many terms it has and its common difference.

SOLUTION

$\frac{n}{2}(5 + 45) = 400$ gives $n = 16$.

$5 + 15d = 45$ gives $d = \frac{8}{3}$.

Answer: 16 terms, $d = \frac{8}{3}$

Q6 An AP has first term 17, last term 350 and common difference 9. Find the number of terms and their sum.

SOLUTION

$17 + 9(n-1) = 350$ gives $n - 1 = 37$, so $n = 38$.

$S = \frac{38}{2}(17 + 350) = 19 \times 367 = 6973.$

Answer: 38 terms, sum 6973

Q7 The common difference of an AP is 7 and its 22nd term is 149. Find the sum of its first 22 terms.

SOLUTION

$a + 21 \times 7 = 149$ gives $a = 2$.

$S_{22} = \frac{22}{2}(2 + 149) = 1661.$

Answer: 1661

Q8 The second and third terms of an AP are 14 and 18. Find the sum of its first 51 terms.

SOLUTION

$d = 18 - 14 = 4$ and $a = 14 - 4 = 10$.

$S_{51} = \frac{51}{2}[20 + 50 \times 4] = \frac{51}{2} \times 220 = 5610.$

Answer: 5610

Q9 The first 7 terms of an AP add up to 49 and the first 17 terms add up to 289. Find the sum of the first n terms.

SOLUTION

$\frac{7}{2}(2a + 6d) = 49$ gives $a + 3d = 7$. $\frac{17}{2}(2a + 16d) = 289$ gives $a + 8d = 17$.

Subtracting, $5d = 10$, so $d = 2$ and $a = 1$.

$S_n = \frac{n}{2}[2 + 2(n-1)] = n^2.$

Answer: $S_n = n^2$

Q10 Show that $a_1, a_2, \ldots$ form an AP when (i) $a_n = 3 + 4n$ (ii) $a_n = 9 - 5n$. Find the sum of the first 15 terms in each case.

SOLUTION

(i) $a_{n+1} - a_n = [3 + 4(n+1)] - [3 + 4n] = 4$, which is the same for every n, so the terms form an AP with $a = 7$ and $d = 4$. $S_{15} = \frac{15}{2}[14 + 14 \times 4] = 525.$

(ii) $a_{n+1} - a_n = [9 - 5(n+1)] - [9 - 5n] = -5$, so the terms form an AP with $a = 4$ and $d = -5$.

$S_{15} = \frac{15}{2}[8 + 14 \times (-5)] = -465.$

Answer: (i) 525 (ii) -465

Q11 The sum of the first n terms of an AP is $4n - n^2$. Find the first term, the sum of the first two terms, the second term, and the 3rd, 10th and nth terms.

SOLUTION

$S_1 = 4 - 1 = 3$, so the first term is 3. $S_2 = 8 - 4 = 4$, so $a_2 = S_2 - S_1 = 1$.

$S_3 = 12 - 9 = 3$, so $a_3 = S_3 - S_2 = -1$.

$a_{10} = S_{10} - S_9 = (40 - 100) - (36 - 81) = -15$.

$a_n = S_n - S_{n-1} = (4n - n^2) - [4(n-1) - (n-1)^2] = 5 - 2n$.

Answer: $a_1 = 3$, $S_2 = 4$, $a_2 = 1$, $a_3 = -1$, $a_{10} = -15$, $a_n = 5 - 2n$

Q12 What is the total of the first 40 positive multiples of 6?

SOLUTION

They are $6, 12, 18, \ldots$ with $a = 6$, $d = 6$: $S_{40} = \frac{40}{2}[12 + 39 \times 6] = 20 \times 246 = 4920$.

Answer: 4920

Q13 What is the total of the first 15 multiples of 8?

SOLUTION

$8 + 16 + \cdots + 120 = \frac{15}{2}(8 + 120) = 960$.

Answer: 960

Q14 Add up all the odd numbers from 1 to 49.

SOLUTION

They are $1, 3, 5, \ldots, 49$, which is 25 terms. $S = \frac{25}{2}(1 + 49) = 625$.

Answer: 625

Q15 A contract fixes a penalty for delay: ₹200 for the first day, ₹250 for the second, ₹300 for the third, and so on, each day ₹50 more than the day before. What is the penalty for a delay of 30 days?

SOLUTION

The penalties form an AP with $a = 200$ and $d = 50$.

$S_{30} = \frac{30}{2}[400 + 29 \times 50] = 15 \times 1850 = 27750$.

Answer: ₹27,750

MathSetu

Q16 ₹700 is to be given as seven cash prizes, each ₹20 less than the one before. Find the value of each prize.

SOLUTION

Let the first prize be ₹a, with $d = -20$. $\frac{7}{2}[2a + 6 \times (-20)] = 700$ gives $2a - 120 = 200$, so $a = 160$.

Answer: ₹160, ₹140, ₹120, ₹100, ₹80, ₹60 and ₹40

Q17 Each section of each class plants as many trees as its class number, from Class 1 to Class 12, and every class has three sections. How many trees are planted?

SOLUTION

One section of each class together plants $1 + 2 + \cdots + 12 = \frac{12}{2}(1 + 12) = 78$ trees.

With three sections of each class: $3 \times 78 = 234$.

Answer: 234 trees

Q18 A spiral is made of 13 consecutive semicircles of radii 0.5 cm, 1.0 cm, 1.5 cm, 2.0 cm, … Find its total length. (Take $\pi = \frac{22}{7}$.)

SOLUTION

A semicircle of radius r has length πr, so the total length is π times the sum of the 13 radii.

Sum of radii $= \frac{13}{2}[2 \times 0.5 + 12 \times 0.5] = \frac{13}{2} \times 7 = 45.5$ cm.

Total length $= \frac{22}{7} \times 45.5 = 143$ cm.

Answer: 143 cm

Q19 200 logs are stacked with 20 logs in the bottom row, 19 in the next, 18 in the next and so on. In how many rows are they placed, and how many logs are in the top row?

SOLUTION

$\frac{n}{2}[40 - (n - 1)] = 200$ gives $n^2 - 41n + 400 = 0$, so $(n - 16)(n - 25) = 0$.

For $n = 25$ the top row would have $20 - 24 = -4$ logs, which is impossible. So $n = 16$, and the top row has $20 - 15 = 5$ logs.

Answer: 16 rows, with 5 logs in the top row

MathSetu

Q20 In a potato race, a bucket is 5 m from the first potato and the 10 potatoes are 3 m apart in a straight line. A runner starts at the bucket, fetches one potato at a time and drops it in the bucket. What total distance does the runner cover?

SOLUTION

For each potato the runner goes there and back. The distances are 2×5, 2×8, 2×11, ..., that is $10, 16, 22, \ldots$ for 10 potatoes.

$S_{10} = \frac{10}{2}[20 + 9 \times 6] = 5 \times 74 = 370.$

Answer: 370 m

4 Exercise 5.4 (Optional)

Q1 In the AP $121, 117, 113, \ldots$, which is the first term to be negative?

SOLUTION

$a_n = 121 - 4(n-1) < 0$ gives $n - 1 > 30.25$, so $n > 31.25$.

The smallest such whole number is 32. Check: $a_{31} = 1$ and $a_{32} = -3$.

Answer: the 32nd term

Q2 In an AP, the 3rd and 7th terms add up to 6 and multiply to 8. Find the sum of its first sixteen terms.

SOLUTION

$(a + 2d) + (a + 6d) = 6$ gives $a + 4d = 3$. So $a_3 = 3 - 2d$ and $a_7 = 3 + 2d$.

$(3 - 2d)(3 + 2d) = 8$ gives $9 - 4d^2 = 8$, so $d = \pm\frac{1}{2}$.

If $d = \frac{1}{2}$, then $a = 1$ and $S_{16} = 8\,[2 + 15 \times \frac{1}{2}] = 76$. If $d = -\frac{1}{2}$, then $a = 5$ and $S_{16} = 8\,[10 - 15 \times \frac{1}{2}] = 20$.

Answer: 76 or 20

Q3 A ladder has rungs 25 cm apart. The rungs decrease uniformly in length from 45 cm at the bottom to 25 cm at the top, and the top and bottom rungs are $2\frac{1}{2}$ m apart. What length of wood is needed for the rungs?

SOLUTION

The gap of 250 cm is made of spaces of 25 cm, so there are $\frac{250}{25} = 10$ spaces and 11 rungs.

The lengths form an AP from 45 to 25 with 11 terms: $S = \frac{11}{2}(45 + 25) = 385$ cm.

Answer: 385 cm (3.85 m)

Q4 Houses in a row are numbered from 1 to 49. Show that there is a house number x such that the sum of the numbers before it equals the sum of the numbers after it, and find x.

SOLUTION

Sum of the numbers before x: $S_{x-1} = \dfrac{(x-1)x}{2}$. Sum after x: $S_{49} - S_x = 1225 - \dfrac{x(x+1)}{2}$.

Setting them equal: $\dfrac{x^2 - x}{2} + \dfrac{x^2 + x}{2} = 1225$, so $x^2 = 1225$ and $x = 35$.

Since 35 is a whole number between 1 and 49, such a house exists.

Answer: $x = 35$

Q5 A terrace has 15 steps, each 50 m long, with a rise of $\frac{1}{4}$ m and a tread of $\frac{1}{2}$ m. Find the total volume of concrete needed.

SOLUTION

The kth step stands $\dfrac{k}{4}$ m high, so its volume is $\dfrac{k}{4} \times \dfrac{1}{2} \times 50 = \dfrac{25k}{4}$ m^3.

Total volume $= \dfrac{25}{4}(1 + 2 + \cdots + 15) = \dfrac{25}{4} \times 120 = 750$ m^3.

Answer: 750 m^3

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